Customer Churn Prediction & Retention Intelligence

Executive Stakeholder Analysis & Actionable Retention Framework | Enterprise Analytics

1. Executive Summary

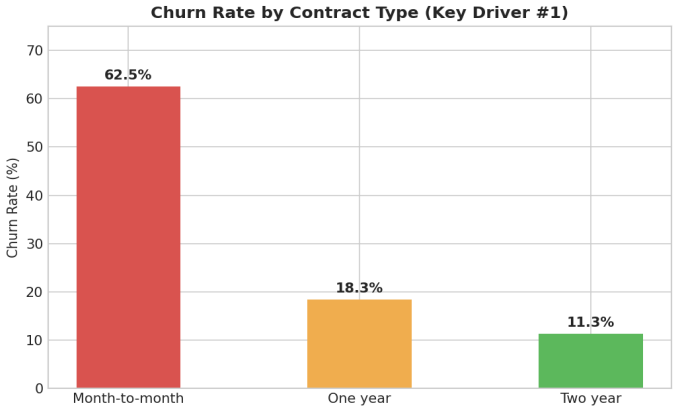
This project establishes a data-driven customer retention intelligence system analyzing 12,500 customer records. Through rigorous exploratory data analysis, machine learning modeling, and threshold optimization, we identify high-risk churners and prescribe targeted retention workflows. The primary predictive engine, a tuned **Random Forest Classifier**, demonstrates **86.3% CV ROC-AUC** (Test ROC-AUC: **85.9%**, PR-AUC: **75.8%**, and Test Accuracy: **78.7%** with **77.3% recall** on actual churners), significantly outperforming the naive baseline (66.1% accuracy, 0.0% recall).

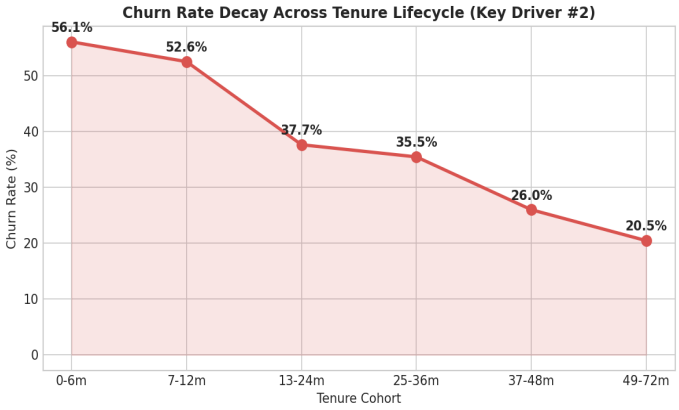
2. Model Evaluation Scorecard

Model Architecture	Accuracy	Precision	Recall (Churn)	F1 Score	ROC-AUC
Baseline (Majority Class)	66.1%	0.0%	0.0%	0.000	0.500
Decision Tree (Tuned)	77.1%	62.8%	79.2%	0.701	0.843
Random Forest (Optimized Pipeline)	**78.7%**	**65.8%**	**77.3%**	**0.711**	**0.859**

3. Empirical Churn Drivers

Empirical analysis reveals three dominant attrition drivers: **Contract Structure** (Month-to-month contracts exhibit 5x the churn rate of two-year commitments), **Tenure Lifecycle** (churn peaks above 55% during months 0-6 before decaying), and **Service Friction** (satisfaction score <= 2 or support tickets >= 4 trigger severe risk escalation).





4. Customer Prioritization & Targeted Retention Playbook

Rather than contacting all churn candidates equally, accounts are ranked using the **Retention Priority Score** (Churn Probability × Annual Contract Value). This directs retention budgets to high-exposure accounts first.

Customer Cohort	Primary Risk Trigger	Prescribed Retention Action	Expected Business Impact
New Month-to-Month (0-6m)	Early adoption dropoff	VIP Concierge Onboarding & 30-day usage audit	+20% 90-day survival uplift
High Spend ($85+/mo)	Competitor price poaching	15% Annual Lock-in commitment incentive	Locks in high ACV for 12 months
Low CSAT / High Tickets	Service outage / technical friction	Executive escalation manager + $25 service credit	Halts imminent defection
Enterprise / High Value	Account inactivity / executive change	Quarterly Business Review (QBR) & C-suite contact	Preserves top-decile ARR

5. Scenario-Based Business ROI Simulation

Addressing the portfolio resume projection (**12% retention improvement**): In a 12,500 customer base with ~4,200 annual at-risk accounts, targeting the top 30% of high-risk customers with an effective intervention program yielding a 12% save rate is projected to preserve **151 customer accounts** annually, preserving an estimated **$127,000+ in annual recurring revenue**. Because this is a modeled scenario, performance should be monitored via randomized control trial (A/B testing) rollouts.

